

2007
HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

MATHEMATICS

General Instructions

- Reading Time – 5 minutes
- Working Time – 3 hours
- Write using blue or black pen
- Board approved calculators may be used
- Write your name on each page
- Each question is to be started on a new page.
- This examination paper must NOT be removed from the examination room

- There is a total of ten questions.
- Each question is worth 12 marks.
- Marks may be deducted for careless or badly arranged work.

Question One

Marks

- | | |
|---|---|
| a) Evaluate $e^{1.4}$ correct to 3 significant figures. | 2 |
| b) Factorise $2x^2 + 7x - 4$ | 2 |
| c) Simplify $\tan 30^\circ \cos 60^\circ$ leaving your answer in surd form. | 2 |
| d) Differentiate $x^2 \ln x$. | 2 |
| e) Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$ | 2 |
| f) Solve $(x+2)^2 = 9$. | 2 |

Question Two (Start a new page)

Marks

a) Solve $3x^2 - x - 5 = 0$ leaving your answer in surd form.

2

b) Differentiate $(4x^3 - 5)^6$

2

c) Find a primitive of $x^2 + \cos 3x$

2

d) Evaluate $\int_0^1 e^{2x} dx$

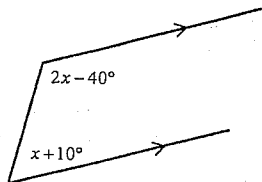
2

e) Given that $\frac{12}{\sqrt{6}} = \sqrt{a}$, find the value of 'a'.

2

f) Find the value of 'x' giving a reason for your answer.

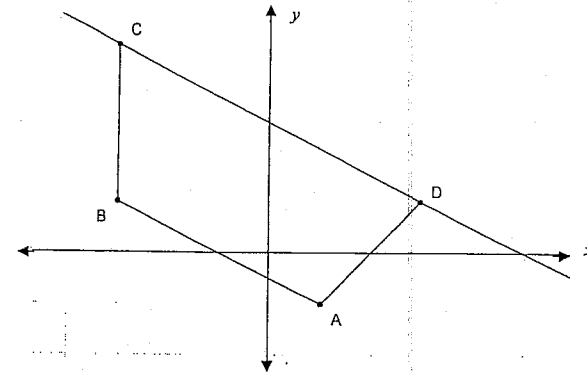
2



Question Three (Start a new page)

Marks

a) A (1, -1), B (-3, 1), C (-3, 4) and D (3, 1) are points on the Cartesian Plane.



i) Find the distance CD.

1

ii) Show that the equation of the line CD is $x + 2y - 5 = 0$.

2

iii) Find the perpendicular distance of A from CD.

2

iv) Hence or otherwise find the area of the triangle ACD.

1

v) What type of quadrilateral is ABCD? Explain carefully.

2

b) Simplify fully $\frac{\sin(180 - \theta) \times \cot \theta}{\sec \theta}$.

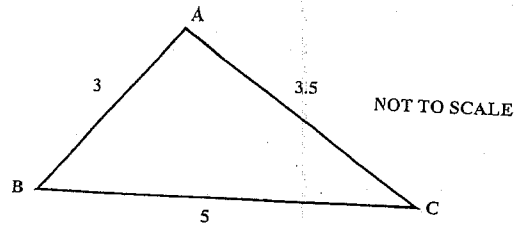
2

c) Factorise and simplify $\frac{8x^3 - 27}{4x - 6}$.

2

Question Four (Start a new page)

a)



In the diagram above $AB = 3\text{cm}$, $BC = 5\text{cm}$ and $AC = 3.5\text{cm}$. Find the size of the smallest angle, correct to the nearest degree.

Marks

2

b) Find the equation of the tangent to the curve $y = e^x + x$ at the point where $x = 0$.

3

c) State the centre and radius of the circle with equation $(x+3)^2 + y^2 = 16$

2

d) i) Sketch on the same diagram.

$y = |x-2|$ and $y = 2x$, showing the 'x' and 'y' intercepts.

2

ii) Hence or otherwise solve $|x-2| = 2x$

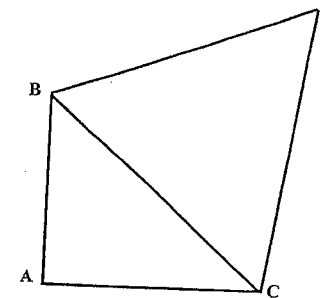
2

iii) Using (i) or otherwise, find $\int_0^4 |x-2| dx$

1

Question Five (Start a new page)

a)



In the above diagram, ABC is an isosceles triangle in which $\angle BAC = 90^\circ$. BCD is an equilateral triangle.

Copy the diagram onto your answer sheet and mark in the given information.

i) Find the size of $\angle ACD$ giving reasons.

2

ii) If $BC = 3\text{cm}$, find the perimeter of ABDC in exact form.

2

b) The quadratic equation $2x^2 - 3x + 6 = 0$ has roots α and β . Find the value of:

i) $\alpha + \beta$

1

ii) $\alpha\beta$

1

iii) $\alpha^2 + \beta^2$

2

c) Use Simpson's rule with 3 function values to find an approximation for the value of $\int_0^1 10^x dx$. Give your answer to 3 decimal places.

2

d) Evaluate $\int_0^{\frac{\pi}{3}} \sec^2 x dx$. Give your answer in exact form.

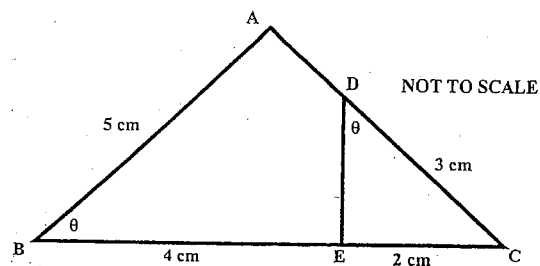
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Marks

Question Six(Start a new page)

Marks

a)



In the above diagram, ABC and DEC are two triangles in which $\angle ABC = \angle EDC$. Also $AB = 5\text{ cm}$, $BE = 4\text{ cm}$, $EC = 2\text{ cm}$ and $CD = 3\text{ cm}$.

Copy the diagram onto your answer sheet.

- Prove that the two triangles are similar.
- Hence, giving reasons find the length of DE .

2

2

- b) Farmer Brown has hired a driller to drill a borehole to gain access to the underground water on his property. The cost is \$260 for the first 3 metres drilled, \$280 for the next 2 metres, \$300 for the next 2 metres and so on. The price increases by the same amount for each successive 2 metres drilled.

- Show that the cost of drilling the portion from a depth of 25 metres to 27 metres is \$500.
- Calculate the total cost of drilling to a depth of 27 metres.
- The cost of drilling the borehole to reach water was \$12500. Find the total depth drilled to give access to the water.
[To gain full marks all working needs to be shown.]

2

1

3

- c) Solve for x : $\log_a 3 = 2\log_a 6 - \log_a x$

2

Question Seven(Start a new page)

Marks

- a) A function $f(x)$ is defined by $f(x) = x^3 - 3x^2 - 9x$.

- Find $f'(x)$ and $f''(x)$.
- Find the turning points for the curve and determine their nature.
- Show that there is one point of inflexion and find its coordinates.
- Sketch the graph of $y = f(x)$ showing the turning points and the point of inflexion.
- Find the values of ' x ' for which the function $f(x)$ is decreasing.

1

3

2

2

1

- b) For what values of ' k ' does the quadratic equation $x^2 - (k+3)x + 4k = 0$

- have one root equal to 2?
- have no real roots?

1

2

Question Eight(Start a new page)

Marks

- a) A particle is moving in a straight line and its velocity v metres/second at time t seconds is given by:

$$v = \frac{dx}{dt} = 1 - 2 \sin 2t, \quad t \geq 0$$

Initially the particle is at the origin.

- Express the displacement x , as a function of t .
- Find the position of the particle when $t = \frac{\pi}{6}$.
- Find an expression for the acceleration in terms of t .
- Sketch the graph of the acceleration for $0 \leq t \leq \pi$.
- What is the maximum acceleration of the particle?

2

1

1

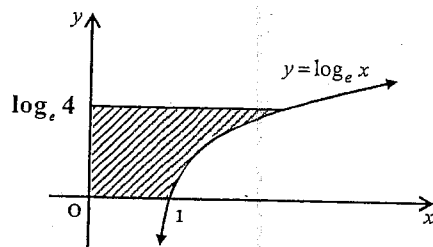
2

1

- b) In the diagram below, the shaded region bounded by the curve $y = \log_e x$, the x and y axes and the line $y = \log_e 4$ is rotated about the y -axis.

3

Find the exact volume of the solid of revolution formed.



- c) The geometric series $1 - x + x^2 - x^3 + \dots$ has a limiting sum of 4.

2

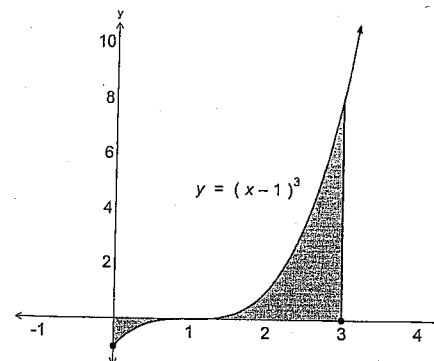
Find the value of x .

Question Nine(Start a new page)

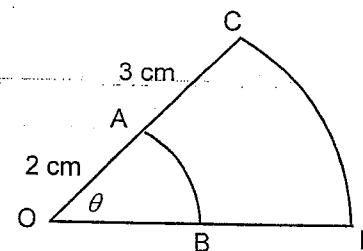
Marks

- a) The shaded area in the diagram below is the region bounded by the curve $y = (x-1)^3$, the x and y axes and the line $x = 3$. Find the shaded area.

4



b)



The arcs AB and CD are parts of concentric circles with centre O. OA = 2 centimetres and AC = 3 centimetres.

- Find an expression for the area of the sector AOB.
- Find the ratio of the area of sector AOB : the area of ABDC

1

2

Question 9 continued over/

Question 9 continued

- c) An industrial plant produces vacuum cleaners. The annual production, P cleaners, at time t years, is given by:

$$P = P_0 e^{kt} \text{ where } P_0 \text{ and } k \text{ are constants}$$

Initially the production of the plant was 2500 cleaners per annum. Five years later it had increased to 4000 cleaners per annum.

- Find the values of P_0 and k .
- What is the predicted production after 10 years?
- Find the rate of increase in production after 5 years.

Marks

3

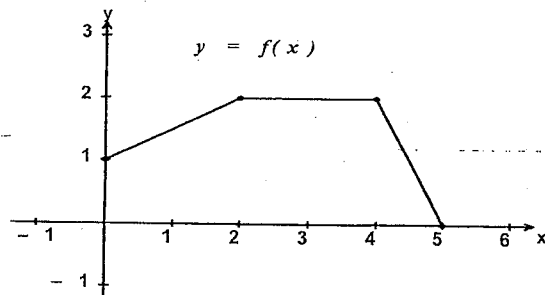
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1

Question Ten(Start a new page)

- a) The graph of $y = f(x)$ is drawn below.

On a number plane sketch the graph of the derivative function $y = f'(x)$.



2

- b) Let A be the point $(-2, 0)$ and B be the point $(6, 0)$.

At the point $P(x, y)$, PA meets PB at right angles.

- Show that the gradient of PA is $m_1 = \frac{y}{x+2}$.
- Hence find an equation for the locus of P.

1

2

Question 10 continued over/

Question 10 continued

Marks

- c) i) Given that $\frac{x^2}{4} + y^2 = 1$,

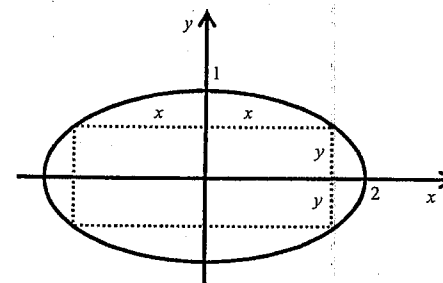
Show that $y = \frac{\sqrt{4-x^2}}{2}$ for $y \geq 0$

1

- ii) Show that $\frac{dy}{dx} = \frac{-x}{2\sqrt{4-x^2}}$

1

- d) i) The ellipse with equation $\frac{x^2}{4} + y^2 = 1$ is drawn below.



A rectangle of length $2x$ and width $2y$ is to be constructed inside the ellipse with its vertices on the ellipse as shown.

Using part (c) or otherwise show that an expression for the area of the rectangle is given by $A = 2x\sqrt{4-x^2}$.

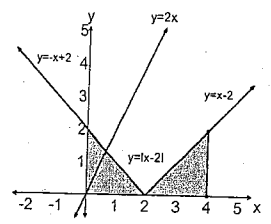
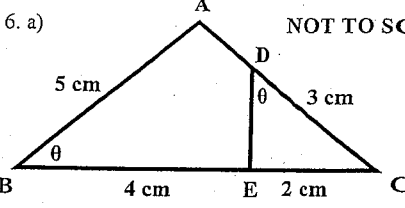
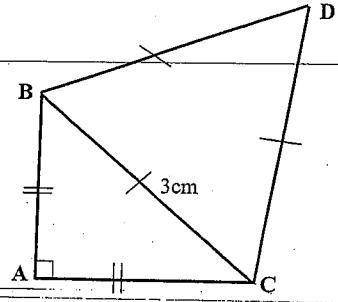
1

- ii) Hence find the value of x so that the area of the rectangle is a maximum.

4

END OF EXAM

CHS 2007 Mathematics (2U) YR 12	HSC SOLUTIONS
1.a) $4.05519967 = 4.06$ {3 Sig Fig} ☒	3.a)i) $CD = \sqrt{6^2 + (-3)^2} = \sqrt{45} = 3\sqrt{5}$ ☒
b) $(2x-1)(x+4)$ ☒	ii) $m_{CD} = \frac{-3}{6} = -\frac{1}{2}$ ☒
c) $\frac{1}{\sqrt{3}} \times \frac{1}{2} = \frac{1}{2\sqrt{3}}$ ☒	$\therefore \text{eqn } CD: y-1 = -\frac{1}{2}(x-3)$ ☒
d) $x^2 \times \frac{1}{x} + 2x \times \ln x = x + 2x \ln x$ ☒	$2y-2 = -x+3 \Rightarrow x+2y-5=0$
e) $\lim_{x \rightarrow 3} \frac{(x-3)(x+3)}{x-3} = \lim_{x \rightarrow 3} (x+3) = 6$ ☒	iii) $a=1, b=2, c=-5, x_1=1, y_1=-1$ ☒
f) $x+2=\pm 3 \rightarrow x=1 \text{ and } x=-5$ ☒	$\therefore d = \frac{ 1 \times 1 + 2 \times -1 + (-5) }{\sqrt{1^2 + 2^2}} = \frac{6}{\sqrt{5}}$ ☒
2.a) $x = \frac{1 \pm \sqrt{1^2 - 4 \times 3 \times -5}}{2 \times 3} = \frac{1 \pm \sqrt{61}}{6}$ ☒	iv) $A = \frac{1}{2} \times \frac{6}{\sqrt{5}} \times 3\sqrt{5} = 9u^2$ ☒
b) $6(4x^3 - 5)^5 \times 12x^2$ ☒ $= 72x^2 (4x^3 - 5)^5$ ☒	v) $m_{AB} = \frac{1 - (-1)}{-3 - 1} = -\frac{1}{2}$
c) $\frac{x^3}{3} + \frac{1}{3} \sin 3x$ ☒	Hence using (ii) $AB \parallel CD$ ☒ So ABCD is a Trapezium ☒
d) $\int_0^1 e^{2x} dx = \frac{1}{2} e^{2x} \Big _0^1$ ☒ $= \frac{1}{2}(e^2 - e^0) = \frac{1}{2}(e^2 - 1)$ ☒	b) $\sin \theta \times \frac{\cos \theta}{\sin \theta} \times \cos \theta = \cos^2 \theta$ ☒
e) $= \frac{12}{\sqrt{6}} \times \frac{\sqrt{6}}{\sqrt{6}} = 2\sqrt{6}$ ☒ $= \sqrt{24} \Rightarrow a = 24$ ☒	c) $\frac{(2x-3)(4x^2+6x+9)}{2(2x-3)}$ ☒ $= \frac{4x^2+6x+9}{2}$ ☒
f) $(2x-40) + (x+10) = 180$ { Co-interior $\angle$'s and $\parallel$ lines } ☒ $3x = 210 \Rightarrow x = 70^\circ$ ☒	4. a) $\cos \theta = \frac{3.5^2 + 5^2 - 3^2}{2 \times 3.5 \times 5} \Rightarrow \theta \approx 36^\circ$ ☒
	b) $y' = e^x + 1$ ☒ when $x=0, m = y' = 2, y = 1$ ☒ $\therefore y-1 = 2(x-0) \Rightarrow y = 2x+1$ ☒
	c) Centre is $(-3, 0)$ and Radius is 4. ☒

CHS 2007 Mathematics (2U) YR 12	TRIAL HSC SOLUTIONS
4.d) i)  ☒	c) $\int_0^1 10^{-x} dx = \frac{1-0}{6} \left[f(0) + 4f\left(\frac{1}{2}\right) + f(1) \right]$ ☒ $= \frac{1}{6} [1 + 4 \times 3.16228 + 10] \approx 3.942$ ☒
ii) $2x = -x + 2$ ☒ $3x = 2 \Rightarrow x = \frac{2}{3}$ ☒	d) $\int_0^{\frac{\pi}{3}} \sec^2 x dx = \left[\tan x \right]_0^{\frac{\pi}{3}}$ ☒ $= \tan \frac{\pi}{3} - \tan 0 = \sqrt{3}$ ☒
iii) Shaded area $= \frac{1}{2}(2 \times 2) + \frac{1}{2}(2 \times 2) = 4$ ☒	6. a)  ☒
5.a) 	i) In the Δ 's ABC and EDC ☒ $\angle ABC = \angle EDC$ { given } ☒ $\angle ACB = \angle ECD$ { common } ☒ $\therefore \Delta ABC \parallel \Delta EDC$ { equiangular } ☒
	ii) $\frac{DE}{AB} = \frac{DC}{BC}$ {corres. sides in similar Δ 's} ☒ $\therefore \frac{DE}{5} = \frac{3}{6} \Rightarrow DE = 2.5 \text{ cm}$ ☒
i) $\angle ACB = \frac{180-90}{2} = 45^\circ$ { $\angle$ sum isos Δ } ☒ $\angle BCD = 60^\circ$ { equilateral Δ } ☒ $\therefore \angle ACD = 45^\circ + 60^\circ = 105^\circ$ ☒	b) Set up 2 arithmetic sequences: Depth: 3, 5, 7, ..., 25, 27, ... ① Cost: 260, 280, 300, ... ②
ii) Let $AB = AC = x; \therefore x^2 + x^2 = 3^2$ ☒ $\therefore 2x^2 = 9 \Rightarrow x = \frac{3}{\sqrt{2}}$ ☒ $\therefore P = 2 \times \frac{3}{\sqrt{2}} + 2 \times 3 = \frac{6}{\sqrt{2}} + 6 \text{ cm}$ ☒	i) Using ①: $a = 3, d = 2; T_n = a + (n-1)d$ ☒ $\therefore 27 = 3 + (n-1) \times 2 \Rightarrow n = 13$ ☒ Using ②: $a = 260, d = 20$ and $n = 13$ ☒ $T_{13} = 260 + 12 \times 20 = \500 ☒
b) i) $\alpha + \beta = \frac{3}{2}$ ☒ ii) $\alpha\beta = 3$ ☒ iii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$ ☒ $= \left(\frac{3}{2}\right)^2 - 2 \times 3 = -3\frac{3}{4}$ ☒	ii) $S_{13} = \frac{13}{2} [260 + 500] = \4940 ☒ iii) $12500 = \frac{n}{2} [2 \times 260 + (n-1) \times 20]$ ☒ $n^2 + 25n - 1250 = 0$ {after simplifying} ☒ $(n-25)(n+50) = 0 \Rightarrow n = 25$ ☒ $\therefore T_{25} = 3 + 24 \times 2 = 51 \text{ metres drilled}$ ☒

6.c) $\log_a 3 = \log_a 6^2 - \log_a x$

$$\log_a 3 = \log_a \frac{36}{x} \Rightarrow x = 12 \quad \checkmark \checkmark$$

7.i) $f(x) = x^3 - 3x^2 - 9x$

$$f'(x) = 3x^2 - 6x - 9; f''(x) = 6x - 6 \quad \checkmark$$

ii) For turning points $f'(x) = 0$

$$\therefore 3(x^2 - 6x - 9) = 0$$

$$3(x-3)(x+1) = 0$$

$$\therefore x = -1 \text{ and } x = 3 \quad \checkmark$$

When $x = 3$: $y = -7, y'' = 12 > 0$

$$\therefore (3, -7) \text{ is a minimum turning point.} \quad \checkmark$$

When $x = -1$: $y = 5, y'' = -12 < 0$

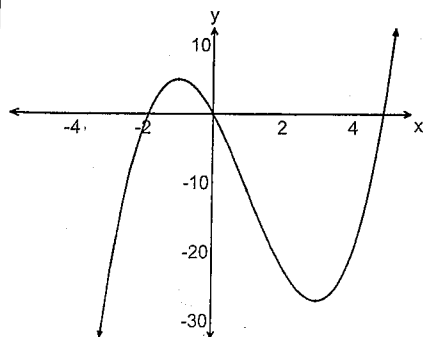
$$\therefore (-1, 5) \text{ is a maximum turning point.} \quad \checkmark$$

iii) For inflexion points $f''(x) = 0$

$$\therefore 6x - 6 = 0 \Rightarrow x = 1 \text{ (Only one point)} \quad \checkmark$$

It has coordinates $(1, -11)$. $\checkmark$

iv) $\checkmark \checkmark$



v) For a decreasing function $f'(x) < 0$

Hence from the graph $-1 < x < 3 \quad \checkmark$

b) i) Substitute $x = 2$ to obtain:

$$4 - 2k - 6 + 4k = 0 \Rightarrow k = 1 \quad \checkmark$$

ii) For no real roots $\Delta < 0$, where $\Delta = b^2 - 4ac$

$$\therefore (k+3)^2 - 4 \times 1 \times 4k < 0 \quad \checkmark$$

$$k^2 - 10k + 9 < 0$$

$$(k-9)(k-1) < 0 \Rightarrow 1 < k < 9 \quad \checkmark$$

8. a) i) $v = 1 - 2 \sin 2t$

$$\therefore x = t + \cos 2t + c \quad \checkmark$$

When $t = 0$, $x = 0$

$$\therefore 0 = 0 + \cos 0 + c$$

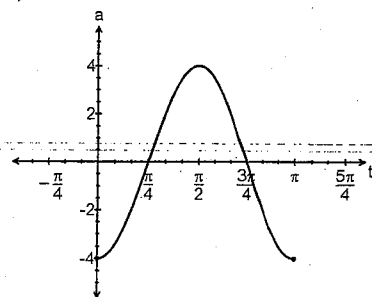
$$0 = 1 + c \Rightarrow c = -1 \quad \checkmark$$

$$\therefore x = t + \cos 2t - 1$$

ii) $x = \frac{\pi}{6} + \cos \frac{\pi}{3} - 1 = \frac{\pi}{6} - \frac{1}{2} \quad \checkmark$

iii) $a = \ddot{x} = -4 \cos 2t \quad \checkmark$

iv)



v) Maximum acceleration is $4 \text{ m/s}^2 \quad \checkmark$

b) $V = \pi \int_0^{\ln 4} x^2 dy \quad [y = \ln x \Rightarrow x = e^y]$

$$V = \pi \int_0^{\ln 4} (e^y)^2 dy = \pi \int_0^{\ln 4} e^{2y} dy \quad \checkmark$$

$$= \frac{\pi}{2} \left[e^{2y} \right]_0^{\ln 4} = \frac{\pi}{2} [e^{2 \ln 4} - e^0] \quad \checkmark$$

$$\frac{\pi}{2} [e^{\ln 16} - 1] = \frac{15\pi}{2} u^3 \quad \checkmark$$

8.c) $S_\infty = \frac{a}{1-r}$, with $a = 1, r = -x \quad \checkmark$

$$\therefore 4 = \frac{1}{1+x}$$

$$4 + 4x = 1 \Rightarrow x = -\frac{3}{4} \quad \checkmark$$

9. a)

$$A = \left| \int_0^1 (x-1)^3 dx \right| + \left| \int_1^3 (x-1)^3 dx \right| \quad \checkmark \checkmark$$

$$= \left| \frac{1}{4} [(x-1)^4]_0^1 \right| + \left| \frac{1}{4} [(x-1)^4]_1^3 \right| \quad \checkmark$$

$$= \frac{1}{4} + 4 = 4\frac{1}{4} \text{ units}^2 \quad \checkmark$$

b) i) $A = \frac{1}{2} r^2 \theta = \frac{1}{2} \times 2^2 \times \theta = 2\theta \quad \checkmark$

ii) Area sector $OCD = \frac{1}{2} \times 5^2 \times \theta = 12.5\theta$

$$\therefore \text{Area } ABCD = 12.5\theta - 2\theta = 10.5\theta \quad \checkmark$$

$$\text{Hence ratio } AOB : ABCD = 2\theta : 10.5\theta = 4:21 \quad \checkmark$$

c) i) $P = P_0 e^{kt}$; when $t = 0, P = 2500$

$$\therefore 2500 = P_0 e^0 \Rightarrow P_0 = 2500 \quad \checkmark$$

Also when $t = 5, P = 4000$

$$\therefore 4000 = 2500 e^{5k} \Rightarrow 1.6 = e^{5k} \quad \checkmark$$

$$\ln(1.6) = 5k \Rightarrow k \approx 0.094 \quad \checkmark$$

ii) $P = 2500 e^{0.094 \times 10} \approx 6400 \quad \checkmark$

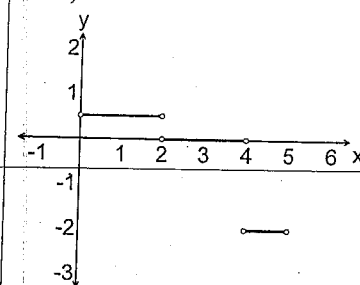
iii) Rate = $\frac{dP}{dt}$: $P = 2500 e^{0.094t}$

$$\frac{dP}{dt} = 0.094 \times 2500 e^{0.094t}$$

$$\text{Hence after 5 years } \frac{dP}{dt} = 0.094 \times 2500 e^{0.094 \times 5}$$

$$\approx 376 \text{ vacuum cleaners / year} \quad \checkmark$$

10. a)



$$\checkmark \checkmark$$

b) i) $m_{PA} = \frac{y-0}{x-(-2)} \Rightarrow m_1 = \frac{y}{x+2} \quad \checkmark$

ii) $m_{PB} = \frac{y-0}{x-6} \Rightarrow m_2 = \frac{y}{x-6}$

Since $PA \perp PB$ then $m_1 \times m_2 = -1 \quad \checkmark$

$$\therefore \frac{y}{x+2} \times \frac{y}{x-6} = -1$$

$$y^2 = -(x+2)(x-6) \quad \checkmark$$

$$\therefore x^2 + y^2 - 4x - 12 = 0$$

10.c) i) $y^2 = 1 - \frac{x^2}{4} = \frac{4-x^2}{4}$

$$\therefore y = \sqrt{\frac{4-x^2}{4}} = \frac{\sqrt{4-x^2}}{2}$$

ii) $y = \frac{1}{2}(4-x^2)^{\frac{1}{2}}$

$$\frac{dy}{dx} = \frac{1}{2} \times \frac{1}{2} \times (4-x^2)^{-\frac{1}{2}} \times -2x$$

$$= \frac{-x}{2\sqrt{4-x^2}}$$

d) i) $A = 2x \times 2y$

$$= 2x \times 2 \times \frac{\sqrt{4-x^2}}{2} = 2x\sqrt{4-x^2}$$

ii) $A = 2x\sqrt{4-x^2}$

$$\frac{dA}{dx} = \sqrt{4-x^2} \times 2 + 2x \times \frac{1}{2}(4-x^2)^{-\frac{1}{2}} \times -2x$$

$$= 2\sqrt{4-x^2} - \frac{2x^2}{\sqrt{4-x^2}}$$

$$= \frac{2(4-x^2)}{\sqrt{4-x^2}} - \frac{2x^2}{\sqrt{4-x^2}}$$

$$= \frac{8-4x^2}{\sqrt{4-x^2}}$$

✓

Stationary points when $\frac{dA}{dx} = 0$

$$\therefore 8 - 4x^2 = 0$$

$$x^2 = 2 \Rightarrow x = \pm \sqrt{2}$$

✓

Since $x > 0$, only need to test $x = +\sqrt{2}$.

x	1.3	$\sqrt{2}$	1.5
A'	0.816	0	-0.756
	/	—	\

✓

✓

$\therefore$ A maximum area occurs when $x = \sqrt{2}$.

✓